2026 Basic Abstract Algebra Study

LECTURE 1 HOMEWORKS

SOLUTIONS

Version
v1.0.0

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Published on
2026-05-21

Hosted by
Coding Lab

Read this carefully before moving on.

Please, clearly specify whether the statement in question is *true* or *false*, for the *prove or disprove* type of questions. As always, it is highly encouraged for you to write each step clearly with full sentences.

As a demonstration of a proof answer, consider the following example.

The above statement is **true**.

Proof. ... □

As a demonstration of a disproof answer, consider the following example.

The above statement is **false**.

Proof. ... □

Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n. For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n. i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{\ldots, -10, -5, 0, 5, 10, \ldots\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B, A is said to be a subset of B if the following condition holds.

If x is an element of A, then x is an element of B.

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A. This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Let A and B be two sets. Let $A(x)$ mean x is an element of A, and $B(x)$ mean x is an element of B.

Which of the following is equivalent to saying that $A \subset B$?

- ○ **1.** $B(x) \to A(x)$
- ○ **2.** not $A(x) \to B(x)$
- ○ **3.** not $A(x) \to$ not $B(x)$
- ○ **4.** $A(x) \to$ not $B(x)$
- ● **5.** not $B(x) \to$ not $A(x)$

A *contrapositive* of a conditional $\mathrm{p} \to \mathrm{q}$ is the conditional not $\mathrm{q} \to$ not p. A contrapositive is *equivalent* to the original conditional. i.e., it is true if and only if the original conditional is true.

By Definition 2, $A \subset B$ means $A(x) \to B(x)$ for any x. From the above choices, the only *equivalent* choice is not $B(x) \to$ not $A(x)$, which is a contrapositive of the original conditional.

Therefore, the fifth choice is the correct answer.

Definition 4 (Roots of unity)

Let μ_n denote the set of all complex solutions to an equation of form $x^n = 1$ for a positive integer n. Elements of such a set are called *n-th roots of unity*.

For instance, $\mu_2 = \{1, -1\}$.

Q2.

EASY

Show that $\mu_2 \subset \mu_4$.

Proof. Let's verify that by manually listing elements of μ_2 and μ_4.

There are two complex solutions to $x^2 = 1$: 1 and -1.

Meanwhile, there are four complex solutions to $x^4 = 1$: $1, -1, i$, and $-i$.

From here, clearly $\mu_2 \subset \mu_4$. □

Now, let's consider squaring each element of μ_2. By the definition, $1^2 = 1$ and $(-1)^2 = 1$. Again, let's square them once more. $(1^2)^2 = 1^2 = 1$, and similarly $((-1)^2)^2 = 1^2 = 1$. Can you find a pattern?

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m.

Hint 5. *Provide concrete n and m that make up a counterexample. Try justifying your answer by using the definition of set builder notation.*

The statement is × **false**.

Proof. Consider $n = 10$ and $m = 8$. Take any odd integer, say, 7. For 7 to be an element of $10\mathbb{Z} + 8\mathbb{Z}$, there should be x, a multiple of 10, and y, a multiple of 8, such that $x + y = 7$. Note that every multiple of 10 or 8 is even. As there are no even numbers that become odd when summed, there are no such x and y. Hence, $7 \notin 10\mathbb{Z} + 8\mathbb{Z}$.

As there exists an element $a = 7$ that is an element of $\mathbb{Z}$ but not an element of $10\mathbb{Z} + 8\mathbb{Z}$, the condition $a \in \mathbb{Z} \to a \in 10\mathbb{Z} + 8\mathbb{Z}$ is false. Thus, $\mathbb{Z} \not\subset 10\mathbb{Z} + 8\mathbb{Z}$. As there is a counterexample in a case where $n = 10$ and $m = 8$, the statement is false. □

HARD

Q4.

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 6. *cf. lecture note.*

The statement is ✓ **true**.

Proof. By Definition 3, we need to show that both $2\mathbb{Z} + 3\mathbb{Z} \subset \mathbb{Z}$ and $\mathbb{Z} \subset 2\mathbb{Z} + 3\mathbb{Z}$ hold.

$2\mathbb{Z} + 3\mathbb{Z} \subset \mathbb{Z}$. Recall that $2\mathbb{Z} + 3\mathbb{Z}$ is a set given by $\{2x + 3y \mid x \in \mathbb{Z}, y \in \mathbb{Z}\}$. If n is an element of $2\mathbb{Z} + 3\mathbb{Z}$, then there exist integers x and y such that $n = 2x + 3y$. From here, we conclude that n is an integer, as $2x$ and $3y$ are integers and their sum is also an integer. (cf. PCSA Q2)

$\mathbb{Z} \subset 2\mathbb{Z} + 3\mathbb{Z}$. Let n be an integer. First, let's try to find two integers x and y such that $1 = 2x + 3y$. For instance, there are $x = 2$ and $y = -1$ such that $1 = 2(2) + 3(-1)$. Now, we multiply both sides by n to get $n = n(2x + 3y) = 2xn + 3yn = 2(xn) + 3(yn)$. As n is a sum of a multiple of 2 and a multiple of 3, it is an element of $2\mathbb{Z} + 3\mathbb{Z}$.

By extensionality, we conclude that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$. □

In fact, any two coprime integers n and m have the same property, i.e., $n \perp m \to n\mathbb{Z} + m\mathbb{Z} = \mathbb{Z}$ for integers n, m. Moreover, not only in $\mathbb{Z}$, but in any *principal ideal domain*, the *ideal* generated by two coprime elements is the entire ring.

Q5.

HARD

Show using Definition 2 that for any two positive integers n and m, $\mu_n \subset \mu_{nm}$.

Hint 7. *Of course, it is a generalization of* ***Q2*** .

Recall the Definition 2 that $\mu_n \subset \mu_{nm}$ essentially means the following.

If x is a solution to $x^n = 1$, then it is also a solution to $x^{nm} = 1$.

Proof. For two positive integers n and m, assume x is an element of μ_n. By the definition, we have $x^n = 1$. By raising both sides to the m-th power, we get $(x^n)^m = x^{nm} = 1^m = 1$. From here, we conclude that x is a solution to the equation $x^{nm} = 1$, thus $x \in \mu_{nm}$.

Therefore, μ_n is a subset of μ_{nm}. □

Q6.

HARD

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n.

Hint 8. *cf. PCSA Q4.*

The following is the same as the solution of PCSA Q4.

Proof.

($\mathbb{Z}_n + \mathbb{Z}_n \subset \mathbb{Z}_{2n-1}$). Let $a, b \in \mathbb{Z}_n$. Then $0 \le a, b \le n-1$, so $0 \le a+b \le 2n-2$, hence $a+b \in \mathbb{Z}_{2n-1}$.

($\mathbb{Z}_{2n-1} \subset \mathbb{Z}_n + \mathbb{Z}_n$). Let $k \in \mathbb{Z}_{2n-1}$, so $0 \le k \le 2n-2$. If $k \le n-1$, write $k = k + 0$ with $k, 0 \in \mathbb{Z}_n$. Otherwise $n \le k \le 2n-2$, and $k = (n-1) + (k-n+1)$ with both summands in $\{0, 1, ..., n-1\} = \mathbb{Z}_n$.

By the Definition 3, $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$. □

Q7.

HARD

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n, then either $n = 1$ or $n = -1$.

Hint 9. *What are the factors of* 1 *and* -1*?*

The statement is ✓ **true**.

Proof. With the Definition 3 in mind, the assumption says that $\mathbb{Z} \subset n\mathbb{Z}$ and $n\mathbb{Z} \subset \mathbb{Z}$ for some integer n. As the latter is pretty obvious, we will focus on the former.

Recall the definition that $n\mathbb{Z}$ is the set of all multiples of n. And by the Definition 2, every integer x should be a multiple of n. If $x = 1$, then 1 should be a multiple of n as well. As the only integer factors of 1 are 1 and -1, n must be either 1 or -1. □

Definition 10 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B.

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cup B \subset (A \cap B) \cup C$.

The statement is ✓ **true**.

Proof.

Forward. Let x be an arbitrary element such that $x \in A \cup B$. We consider two mutually exclusive cases:

(a) **Case 1 ($x \in A \cap B$)**
If x is in the intersection, it trivially follows that $x \in (A \cap B) \cup C$.

(b) **Case 2 ($x \notin A \cap B$)**
Since $x \in A \cup B$ but $x \notin A \cap B$, by the definition of symmetric difference, $x \in A \triangle B$. By our assumption ($A \triangle B \subset C$), this implies $x \in C$. Therefore, $x \in (A \cap B) \cup C$.

Since $x \in (A \cap B) \cup C$ holds true in both cases, we conclude that $A \cup B \subset (A \cap B) \cup C$.

Backward. Let x be an arbitrary element such that $x \in A \triangle B$. By definition of the symmetric difference:

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B$$

By our assumption ($A \cup B \subset (A \cap B) \cup C$), since $x \in A \cup B$, it must hold that:

$$x \in (A \cap B) \cup C$$

This statement means that $x \in A \cap B$ or $x \in C$. However, we already established that $x \notin A \cap B$. Therefore, it must be true that $x \in C$. Consequently, we conclude that $A \triangle B \subset C$. □

Q9.

LEVI

For an integer n, let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P be the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

(a) Show that every two distinct elements of P are disjoint.

Hint 11. *Use Definition 3 and contraposition.*

Proof. Let us rephrase the statement first. The original statement is $M(a) \neq M(b) \to M(a) \cap M(b) = \varnothing$ for all $a, b \in \mathbb{Z}$. Its contrapositive is $M(a) \cap M(b) \neq \varnothing \to M(a) = M(b)$ for all $a, b \in \mathbb{Z}$.

Assume $M(a) \cap M(b) \neq \varnothing$ for some $a, b \in \mathbb{Z}$. As it is not empty, there exists at least one element c in $M(a) \cap M(b)$. By the definition, $c \equiv a \pmod{n}$ and $c \equiv b \pmod{n}$.

If x is an element of $M(a)$, then $x \equiv a \equiv c \equiv b \pmod{n}$ by the symmetry and transitivity of the congruence. Hence, x is an element of $M(b)$. Similarly, if x is an element of $M(b)$, then $x \equiv b \equiv c \equiv a \pmod{n}$, showing that $M(a) = M(b)$ as desired. □

If you're up for a challenge, try showing the symmetry and transitivity of the congruence used here.

The real name of $M(k)$ is *coset*. You just proved that they create a complete *partition* on $\mathbb{Z}$!

(b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Proof. Let a and b be integers.

Forward. If x is an element of $M(a) + M(b)$, there exist two integers y, z such that $y \equiv a \pmod n$ and $z \equiv b \pmod n$ and $x = y + z$. As $x - a - b = (y - a) + (z - b)$ is a multiple of n, $x \in M(a + b)$.

Backward. If x is an element of $M(a + b)$, then $x = k + a + b$ where k is a multiple of n. Then, there exist two integers $k + a$, b such that $k + a \equiv a \pmod n$ and $b \equiv b \pmod n$ and $x = (k + a) + b$.

By extensionality, we conclude that $M(a) + M(b) = M(a + b)$. □

In such a case, we often say that *the operation is well-defined on cosets*. The subgroups for which this coset operation is well-defined are exactly the *normal* ones; in an abelian group such as $\mathbb{Z}$, every subgroup, including $n\mathbb{Z}$, is normal. Moreover, we can now view M as a special kind of function, called a *canonical homomorphism to the quotient*.

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2121} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: 1

As we are to find real solutions, we can think f as a real function. Then, observe that f is strictly increasing, as $x < y$ implies $f(x) < f(y)$.

Now, let S denote the set of all real solutions to the equation $f(x) = x$ and S_{2026} denote the set of all real solutions to the equation $f^{2026}(x) = x$. We will show that $S = S_{2026}$ by the Definition 3.

Proof.

$\boldsymbol{S \subset S_{2026}}$. Let x be an element of S. As $f(x) = x$, we have $f^{2026}(x) = x$, inductively, showing that x is an element of S_{2026}.

$\boldsymbol{S_{2026} \subset S}$. Let x be an element of S_{2026}. If $x < f(x)$, then inductively $x < f^{2026}(x) = x$, which is a contradiction. Similarly, if $x > f(x)$, inductively $x > f^{2026}(x) = x$, which is a contradiction. Therefore, x must be equal to $f(x)$, which amounts to saying that x is an element of S. □

The only real solution to $f(x) = x$ is 1. As S is a singleton, so is S_{2026}.

Therefore, the correct answer is 1.

Q11.

LEVI

Let T be the set $\{z \in \mathbb{C} \mid |z| = 1\}$ and μ_∞ be the set of all roots of unity as follows.

$$\mu_\infty = \bigcup_{n=1}^{\infty} \mu_n = \mu_1 \cup \mu_2 \cup \dots$$

Prove or disprove that $T = \mu_\infty$.

The statement is × **false**.

If you have learned set theory, you might have immediately realized that T is uncountable while μ_∞ is countable.

Here goes the proof!

Proof. Let's first ensure that μ_∞ is countable. We know that each μ_n is finite as $|\mu_n| = n$ and not empty as they always include 1, making μ_∞ not empty. Now take any function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mu_\infty$ such that if $m \le n$, then $f(n, m)$ is the m-th element of μ_n when ordered by arguments. As such f always exists and is clearly surjective, we have $|\mu_\infty| \le |\mathbb{Z}^+ \times \mathbb{Z}^+| = \aleph_0$. Hence, μ_∞ is countable. On the other hand, T is uncountable as the argument $\arg : T \to [0, 2\pi)$ is bijective, and $[0, 2\pi)$ is uncountable. Since one is countable while the other is uncountable, they cannot be equal. □

Note that the above proof is perfectly valid even though it doesn't provide any counterexample. Nevertheless, I provide the below 'counterexample-based' disproof as well.

Proof. Let $x = \frac{4}{5} + \frac{3}{5}i$. It is clear that $|x| = 1$ hence $x \in T$. You may have already realized that this counterexample is from Pythagorean numbers.

If x is an element of μ_∞, there must exist a positive integer n such that $x^n = 1$, otherwise put, $(4 + 3i)^n = 5^n$. Now, let $(4 + 3i)^k = A_k + B_k i$ for some $k \in \mathbb{Z}^+$ and find the next power.

$$(4 + 3i)^{k+1} = (A_k + B_k i)(4 + 3i) = (4A_k - 3B_k) + (3A_k + 4B_k)i$$

Thus, $A_{k+1} = 4A_k - 3B_k$ and $B_{k+1} = 3A_k + 4B_k$. As $A_1 = 4$ and $B_1 = 3$, we can ensure that every A_k, B_k are integers by induction. From here, it is not hard to show that $A_k \equiv 3B_k \pmod 5$ and $B_{k+1} \equiv 3B_k \pmod 5$. As $B_1 = 3$, we can conclude that every B_k is essentially congruent to a power of 3, namely $B_k \equiv 3^k \pmod 5$.

Since 5^n is a real number, its imaginary part, B_n, must be 0 when $k = n$. However, the congruence equation $3^n \equiv 0 \pmod 5$ does not have a solution, as $3 \perp 5$. Thus, such positive integer n does not exist, contradicting with the assumption. Therefore, x cannot be an n-th root of unity for any positive integer n. □

Keep in mind that multiplication of a complex number is a *rotation* on the complex plane. Thus, what we've just shown here is that the angle (*argument*) of x does not divide the circle evenly. Observe, or at least, feel how the problem of evenly dividing a circle turns into a problem of solving a solution to a modular congruence. The roots of unity can be visualized as an exact rotations of a circle, justifying why they are called *cyclic groups*.

Of course, there is a shortcut to the above result. A corollary of the Theorem 12 guarantees that for a rational number other than 0, 1, and -1, its arctan is always an irrational multiple of π. In the above case, $\arctan\left(\frac{3}{4}\right)$ should be an irrational multiple of π, which makes it never reach 1 no matter how many times it rotates.

Theorem 12 (Niven)

If $\theta \in \left[0, \frac{\pi}{2}\right]$ and both $\frac{\theta}{\pi}$ and $\sin\theta$ are rational, then $\sin\theta$ is either 0, 1, or $\frac{1}{2}$.

There is also a way more 'algebraic' proof as well.

Proof. Let $x = \frac{4}{5} + \frac{3}{5}i$ and claim it's an n-th root of unity for some $n \in \mathbb{Z}^+$. Hence we get $(4+3i)^n = 5^n$. In the Gaussian integers, both sides factor into primes: since $(2-i)^2 = 3-4i$, we have $4+3i = i(2-i)^2$, while $5 = (2+i)(2-i)$. Raising each to the n-th power gives the following.

$$i^n(2-i)^{2n} = (2+i)^n(2-i)^n$$

Note that $(2+i)$ and $(2-i)$ are non-associate primes. From here, the right hand side has $(2+i)$ as a prime n times, while the left hand side does not contain it. Recall that a prime factorization by Gaussian integers must be unique up to units. As $n \neq 0$ by the Definition 4, this is a contradiction. Therefore, x is not an n-th root of unity for any positive integer n. $\square$

From here, we have two different sets T and μ_∞. However, as you might have already guessed, μ_∞ is a subset of T. If x is a solution to $x^n = 1$ for some positive integer n, we have $|x^n| = |1|$ by taking absolute values on each side, from which we get $|x^n| = |x|^n = 1$. The only real solution to it is $|x| = 1$.

In fact, μ_∞ forms a *subgroup* of T under multiplication. Moreover, every μ_n is itself a subgroup of μ_∞ and called *cyclic* as every element of μ_n can be written as an exponentiation of *a primitive n-th root of identity*. Finally, T, as you might have imagined, is called the *circle group*, and is also a subgroup of a much larger group $U(\mathbb{C})$, called the *unit group of the complex ring*.

$$\mu_n \leq \mu_\infty \leq T \leq U(\mathbb{C})$$

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

No.	**Evaluation**	**No.**	**Evaluation**
Q1		Q7	
Q2		Q8	
Q3		Q9	
Q4		Q10	
Q5		Q11	
Q6			

There are these additional questions as well; feel free to answer them.

(a) Which was the hardest for you, among those you have solved? __________

(b) Which seemed to be the hardest for you, among those you haven't solved? __________

(c) Which was your favorite question, if you had to choose one? __________